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Global production network mapping for transforming socio-ecological systems

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In the seafood industry, global production networks (GPNs) are so complex that working with individual supply chains to improve sustainability is not enough to create systemic change. A system-level perspective can build upon supply-chain focused seafood certification and scorecard programs that currently dominate the sustainable seafood arena. We present a system-mapping method we piloted with seafood industry leaders, researchers, and experts, designed to visualize individual supply chains in a wider context and generate new ways of looking at old fishery sustainability problems. With simplified schematics of the GPNs of two fisheries, where pressure to transform came from the harvester and the buyer sides, respectively, we show how system maps helped seafood industry participants to locate major gaps in their understanding of the GPN, and to appreciate the extent of their leverage to address persistent problems.

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Introduction

Incremental improvements to single seafood supply chains are insufficient to create transformative change in fisheries globally. Even with aquaculture now accounting for half of the seafood in the marketplace, radical changes in the seafood system and wild capture fishery supply are needed to meet growing global seafood demand [1]. Achieving more sustainable practices in individual seafood supply chains is necessary and positive, but insufficient to transform fisheries systems overall. There is a disconnect between linear, one-directional supply chain management approaches, and today's globalized seafood supply context [2••]. The concept of global production networks (GPNs) better accounts for the relationships among actors and how they are embedded within local and global structures and institutions [3••] — a perspective that is more likely to facilitate the emergence of transformational ideas and actions.

NGOs and government agencies were once the main actors seeking to help seafood harvesters improve fishery sustainability. Today we see increased engagement of private-sector GPN actors, from large seafood companies to start-ups, applying their leverage and using market-based tools to improve harvesting, processing, sourcing and tracking, distribution of and communication about sustainable seafood. Over the past decade many seafood companies' efforts have focused on market-oriented strategies such as eco-label certification through the Marine Stewardship Council (MSC), and scorecards such as Seafood Watch that help buyers avoid sourcing from overfished fisheries [4•].

MSC is a widely recognized brand in seafood eco-certification [5] and as of 2009 seven percent of wild harvested seafood is certified or under review [6]. MSC also plays a role in industry-led Fisheries Improvement Projects (FIPs), which often prepare fisheries for MSC appraisal, or use the MSC sustainability criteria as a guide for improvement [7•]. Both MSC certification and FIPs currently are applied to single fisheries and species, although current system-focused research urges re-integrating species management into whole social–ecological systems [4]. Critiques of MSC certification include that many fisheries and actors in the GPN are excluded because they lack capital, data and organizational structures to participate (as in fisheries that are part of evolving economies [8]), or because their existing sustainability efforts compete with the MSC framework in some way (as in the case of Alaska, which has its own strictly regulated and enforced eco-labeling system). Most significantly from the perspective of resilience and large-scale transformation toward sustainability, evidence shows that incremental improvements at local-fishery scales or in specific markets or countries are insufficient to transform complex global seafood production networks. Single-chain/species/fishery-focused efforts all operate within a global seafood system [9••], and almost every system has leakage [10•]. In other words, cleaning up a fishery or chain of custody in one site may just displace the problem to

another area. For example, non-MSC certified resources have been fraudulently mislabeled as certified [11•,12].

Scorecards used by retailers, distributors, chefs and consumers are based on fishery stock and ecological parameters [4,13]. The goal is to create pressure for fisheries to get onto approved species lists. Scorecards do appear to affect consumer buying behavior [13], despite confusion caused by multiple scoring systems [14] and frequent lack of verification of seafood sourcing. Leakage can occur when buyers opt to blacklist fisheries that do not rise to approved lists in short time scales, and fisheries may have to seek other markets (perhaps with less stringent sustainability standards). The same problem has been documented for commodity species interventions outside of certifications and scorecards. An example is restricting international trade through the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) for sturgeon and sea cucumber [15–17]. Following the CITES listing, sturgeon sales were reduced by 10% [17], and the harvest of some types of sturgeon reduced by half [18]. However, mislabeled or counterfeit sturgeon was still for sale [17,19,20]. Purcell *et al.*'s analysis of the status of sea cucumbers shows that CITES listing alone was not enough to protect luxury seafood items on global markets [15].

Chains within networks

Language used to conceptualize industry-led sustainability actions — namely the pervasive linear supply 'chain' metaphors used in business strategy and academic writings — may hinder our ability to visualize GPNs as complex systems with feedbacks and non-linear relationships within them. Bush *et al.* [2••] reviewed how sustainability governance has been conceptualized in the supply chains and networks literature, and they also provide insight into how corporate social responsibility relates to resource governance, filling spaces not occupied by government, NGOs, and other institutions. In Bush *et al.*'s typology, two modes of sustainability governance include '*in chain*' governance actions within the firm (in our terms, managing the logistics of seafood certification schemes internally) and '*of chains*' inter-firm governance (i.e. leading buyers setting conditions of market access) [2••]. Both of these chain-focused approaches have some elements of 'one-way' flows where a single firm sets the terms. But a third mode, governance '*through chains*', focuses on effecting change beyond the chain [2••], including two-way influences between other firms (even competitors), consumers, and NGOs, among others. Governance through chains recalls concepts of interactive governance, that is, paying attention to these two-way influences of how the agents being governed, for example, also influence governance [21,22••] and new institutional economics and transaction costs, policy networks, polycentric governance, and complex adaptive systems [23••]. That chain thinking may reinforce top-down ways of seeing multi-dimensional problems is highlighted by Mol's analysis of seafood transparency/traceability initiatives, [24••] which can have the same unintended adverse effects as certification and scorecards: all isolate specific chains within a globalized production network and label them as good or bad. This disempowers people working in fisheries and the approach requires increased surveillance and top-down control, without necessarily generating new information about whether environmental performance is improved. A critique by Von Geibler cites palm oil production [25, see Table 3 in the article] to illustrate how the chain approach leaves out parts of the value chain, fails to integrate small-scale farmers, small companies, and consumer stakeholders, and discourages big-picture sustainability vision (e.g. area-wide conservation systems).

To move beyond single supply chain approaches, collaborative tools are needed that bring into the frame more actors, processes, and environments. In the next section we discuss one such tool we developed and piloted with participants representing fisheries and supply networks in Indonesia and Chile.

Methods

We hosted a workshop at the Royal Swedish Academy of Sciences in Stockholm, Sweden, with participants representing fishery cases with diverse geographies, market structures, and socio-ecological contexts (Textbox 1). The workshop preparation and process was designed to facilitate network and system thinking about the cases. We mapped schematic networks to rapidly represent fishery systems and highlight barriers and opportunities for sustainability actions and transformative change. Mapping can clarify the most effective areas and people to engage in a system, referred to as leverage points in systems thinking [26••].

Box 1 Workshop participants from different sectors were central to the co-design process and included:

Head of Sustainability of Denmark-based seafood company Espersen A/B, that led what is considered a successful fishery transformation and the MSC certification of Baltic Cod, by uniting cod processors in the region to extinguish illegal landings [27].

- Co-founder of Pesca en Línea, a fisher-led start-up in Chile that sells artisan-caught seafood direct to chefs and consumers.
- Production network representatives of Indonesia's small-scale blue swimming crab industry including the director of Asosiasi Pengelolaan Rajungan Indonesia (APRI), an association of crab processors, and leaders of the National Fisheries Institute Crab Council ('Crab Council' hereafter).
- The CEO and founder of the NGO Sustainable Fisheries Partnership, a main player in efforts to improve sustainability through Fishery Improvement Projects (FIPs).
- A sustainability consultant experienced in agricultural transformation design.
- Researchers with expertise in social, economic and ecological dimensions of fisheries, system resilience and transformations.

Pre-workshop preparation: The three cases we selected for study included the Baltic Cod fishery that, through actions by seafood processing companies, turned around a near collapse to become MSC certified. The other two cases were two small-scale fisheries in south America and south Asia that were experimenting with approaches to change unsustainable dynamics. We used the Baltic cod case as a reference point for analyzing how fishery transformation can occur. We highlight in this paper the two small-scale fishery examples to show how mapping can reveal opportunities for transformation. Both small-scale fisheries had already invested energy and capital in trying to correct problems and launch new initiatives (see Textbox 2). In the Chilean artisanal mixed-species fishery case, leaders of local fisher organizations and NGO partners were pressing for change. The Indonesian small-scale fishery in this example has no such formal fisher organizations, and pressure for change came from US-based Indonesian blue swimming crab importers seeking to restore declining stocks and ensure sustainable future supply. We asked key individuals associated with the cases to describe (1) the supply chain and its actors, and (2) tried or proposed solutions to date and any outcomes. To elicit system-level thinking we asked questions such as: *Aside from your product, what other fisheries do the processors work with? Do you know whether harvesters capture other products as part of their overall livelihood strategy? What type of gear is used to harvest? Where do people harvest (inshore* vs. *offshore, for example)?* We used the interviews to create preliminary schematics representing all the actors, including competitors and collaborators outside of supply chains, and details about roles and relationships in the system and the environment. When faced with very large numbers of actors of a certain type cluttering the schematic, we simplified them by creating categories. For example, the Indonesian seafood processor sector was simplified to categories such as high volume processors that processed multiple fisheries and low volume processors that processed only crab.

Box 2 Case Descriptions

Blue swimming crab — Indonesia. Pressure for sustainability action stems from the buyer level, as US-based crab importers (member companies in the Crab Council) encourage Indonesian upper tier crab processors to eliminate harvest and sale of illegal, underreported, and unregulated (IUU) crab through the organizational umbrella APRI (see Text Box 1). Most fishery problems and proposed interventions to date had focused on two ends of the supply chain: achieving agreement among competitors at the buyer level, and addressing perceived problems at the harvester level (assumed lack of awareness of catch regulations among fishermen). In terms of Bush *et al.*'s typology [2••], the Crab Council is attempting governance *through the chain*, as each member firm in the Crab Council seeks to clean up its individual supply chain (seeking partnerships with government and NGOs, aiming for MSC certification targets) as well as working together for sustainability at a meta-level of the larger blue swimming crab supply network.

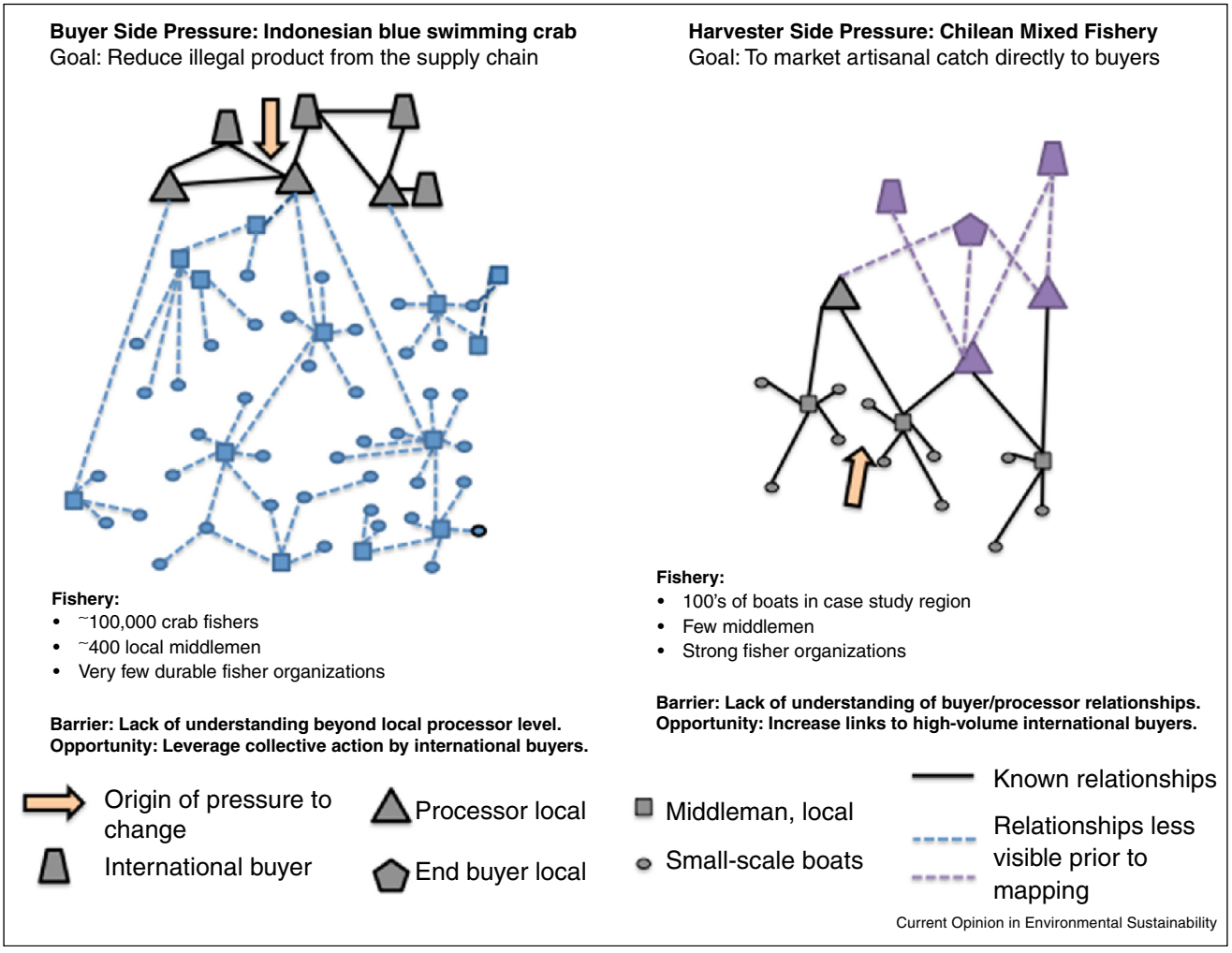
Small-scale mixed species fisheries — southern Chile. Push for change comes from the harvester level, and there is no major seafood industry partner at the buyer level (e.g. a retail chain partner to insist upon and sponsor greater sustainability in the supply). Pesca en Línea, an initiative of the fishing federation Federación Interregional de Pescadores Artesanales del Sur Región de Los Ríos (FIPASUR), has worked to increase prices for products caught by local artisanal fishermen and find a niche market for responsibly harvested seafood. By shortening the supply chain, eliminating traditional middlemen, and marketing outside of the global commodity market, they seek to re-value artisanal catches. Pesca en Línea had won prizes as a Chilean market innovation success, but the scale of operation was small. Strengths in this case include strong fisher networks and resource co-management at local, regional, and national levels. The strategy to advance the artisanal sector includes FIPs, seeking certification for some species, and using MSC standards for improvement even if certification is yet out of reach for other species. This is also an example of governance *through the chain* [2••], as Pesca en Línea works to unite artisanal competitors to add value to artisanal fisher labor and marine resources.

During pre-workshop interviews, critical data verification and communication needs were highlighted by the facilitators. In our case study where pressure for change came from the seafood company side, consensus was lacking regarding certain basic facts such as the number of harvesters in the fishery. Conversely, in our case study where pressure for change came from the harvester side and the workshop participant represented the harvester perspective, knowledge of system actors at points beyond the harvester level was limited.

Schematic mapping of seafood systems during the workshop: First we asked participants to describe: *What are the main problems you are facing? What are some key areas that need improvement? What is happening that you are excited about?* Then we gave cut-out pieces of the preliminary schematic network diagrams that were developed in advance to visualize cases, and we let participants place those actors and elements where they felt they belonged in the system. Participants moved, recombined, and added new cards (elements) as needed. Then we asked the group to identify: *Who/what is missing from the picture? Where are the problem areas? Where are good things or innovation happening in this picture?* We employed mixed breakout groups of supply system 'insiders' and 'outsiders' for each of the cases to ensure fresh perspectives, and expand the knowledge network and the capacity to visually represent a whole system. We re-sorted breakout groups three times so that each case study was talked through with three different sets of participants. Each time the groups rotated to a new fishery case we asked participants questions such as, *What are the differences between this map and the previous one? Where are the opportunities for change in the system and why do you see them as opportunities?*

The final schematic networks included multiple social sectors in a country and sometimes multiple nations in a production network, marine and terrestrial ecosystems,

Figure 1



Simplified schematics of global production networks generated for blue swimming crab in Indonesia (left) and mixed small-scale fisheries of southern Chile (right). Blue and purple regions represent areas of the production network that were less visible to participants prior to the system-mapping workshop.

seafood markets, governance and policies, and interactions and feedbacks among these elements. A simplified GPN is presented in Figure 1.

Results

Mapping the actors and challenges in the systems and sustainability initiatives made missing connections obvious. In the example of change directed from the buyer level in the Indonesian case, there was often conflicting information about the composition, scale and functioning in the fishery and the roles and exact numbers of middlemen and first tier processors (Figure 1, blue region; Table 1, barriers). In the example of change directed from the harvester level in the Chilean case, key information was lacking about processors controlling activity in the middle and about higher volume buyers, especially in international markets and emerging domestic markets where sustainable and artisanal products are valued more highly (Figure 1, purple region; Table 1, barriers).

Opportunities and leverage points emerged through visual mapping (Table 1, opportunities), including the unique structure of small-scale fisher organizations in Chile, and of the Crab Council, itself an innovative model for sustainability governance in the industry. To date the participants from the Chilean case had focused most academic and scientific collaborations on ecological/fishery stock assessments (aiming toward MSC certifications), but now may turn more attention to the social–economic dynamic, and understanding economic *and* non-economic reciprocal and patron–client relationships between producers, buyers and sellers in this system. Leaders from the Crab Council, APRI, and Sustainable Fisheries Partnership developed an idea at the workshop for an

Table 1

Goals, opportunities, and barriers identified through production network mapping

	Indonesian blue swimming crab	Chilean mixed small-scale fisheries
Goal	Eliminate illegally harvested (e.g. undersized) crabs from supply chain.	Create and expand new niche markets for artisanal, sustainable Chilean seafood.
Opportunities	The Crab Council is a noteworthy organization that provides an organizational structure to minimize risk (leakage of unsustainable product) among competitors and maximize sustainability outcomes.	Chilean fisher organizations are strong and well linked with the government and each other. They have potential to successfully rally new niche seafood markets.
Barriers	Most harvesters are not formally organized, making it more challenging to collaborate on sustainability initiatives. The middle of the supply chain (primary processing (including processing sites that are often called ''miniplants'' in this fishery, which may also involve middlemen)) had been repeatedly skipped over in sustainability planning because importers leading the efforts had few network ties in that sector. Changes restricting flow of undersized crabs could result in short term economic losses for the harvesters as well as companies.	Fisher organizations lacked understanding of buyer relationships in the system (who sells what to whom, profit margins at each intermediary level). The focus to date had been slowly developing local domestic markets and seeking ecological fishery certifications, missing opportunities for large volume sustainable seafood markets.
Outcome	Middle/lower processing sectors (rather than harvester sector) was recognized as a leverage point for intervention. Sites where crab is pre-processed locally are the first auditable points in the production network and can become the focus of interventions. Harvesters need to be engaged in planning and communication efforts. Existing community-level institutions could be explored as opportunities to enable better communication regarding fishery sustainability.	Instead of focusing exclusively on ecological assessments and partnerships to certify niche market species, we co-designed collaborative social research questions on relevant seafood markets, actors and incentives, and international market entry points for commodities fished by the artisanal sector, e.g. sardine.

industry-led compliance strategy to eliminate undersized crab from the supply, building on an opportunity provided by the newly enforced fisheries law in Indonesia, with supplier contracts similar to those used in the Baltic Cod case. Remaining questions were how to mitigate short-term negative social and economic effects of the auditing process on harvesters, small primary processors and others, how to clarify economic incentives for network actors such as local middlemen to collaborate, and how to communicate the new strategy in a fishery spanning highly diverse cultural regions and dialects. In addition, because the outcomes (Table 1, Figure 1) highlight what is not well known to the participants co-designing research is a natural next step after mapping exercises are completed.

Using network schematics to visualize supply chains within larger seafood systems helped participants think creatively about future interventions and their likely effects. A workshop participant commented that mapping was 'like seeing the trees, the forest and the people who lived in the forest at the same time.' This type of systemic vision is needed to transform social-ecological seafood systems.

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